

Kerolos Rafeek Farouk Samaan

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SUMMARY

Materials Engineer with experience in mechanical testing, metallurgical analysis, and materials characterization. Skilled in analysis of microstructure, evaluating test results, and preparing technical documents to support quality and engineering requirements. Experienced in working with engineering specifications, material data, and standard documentation in industrial environments.

KEY SKILLS

- Materials Characterization
- Surface Coating Technologies
- Microstructure Analysis
- Non-Destructive Testing (NDT)
- Mechanical Testing & Inspection
- Manufacturing Processes
- Quality Assurance & Control
- Metallurgical Analysis
- Failure Analysis
- Technical Reporting
- AutoCAD, SolidWorks
- Minitab
- CAE analysis
- WAAM, 3D Printing
- Microsoft Office

EMPLOYMENT

ENPPI | Piping Material Engineer

Oct 23, 2025– Present

- Reviewed client specifications, project requirements, and applicable standards to prepare and revise the manual valve datasheet.
- Improved and updated 3+ valve datasheets to ensure alignment with client requirements and material specifications.
- Revised stock codes and material descriptions to maintain alignment with material requisitions and engineering documentation.
- Coordinating with 5-7 suppliers and the procurement team to clarify required components, materials, and technical requirements.
- Supported piping/material engineering activities by preparing and reviewing material specifications, line classes, and material take-offs (MTOs) aligned with project requirements.
- Reviewed vendor datasheets / technical submittals and coordinated comments with discipline teams to ensure compliance with project specs and applicable standards.

INTERNSHIPS EXPERIENCE

Intern | Jushi Group

Aug 7, 2024 – Aug 24, 2024

- Acquired hands-on experience with 4–6 core glass fiber manufacturing processes, including raw material preparation, high-temperature furnace operation, fiber extrusion, cooling, and final product packaging.
- Participated in technical consultations with senior plant personnel, gaining insights into manufacturing system integration, product line management, and packaging logistics.

Intern | United Company for Foundries

Aug 1, 2023 – Sep 15, 2023

- Gained practical exposure to foundry operations, including material selection, casting processes, and quality assurance methodologies within a high-volume industrial environment.
- Executed dimensional and quality control inspections on over 10 cast products, ensuring adherence to engineering design specifications and production tolerances.

CERTIFICATIONS

- NDT Level I & II Certification - Visual Testing, Radiographic Testing, Magnetic Particle Testing, Penetrant Testing, and Ultrasonic Testing | In Progress
- Autodesk Inventor – Online Training Program | Aug 2021
- IELTS 6.5 – British Council | 2021

RESEARCH & PUBLICATIONS

- Samaan, K. R., et al. "[Mechanical and Thermal Performance Optimization of FDM-Printed PETG and PETG-CF Composites.](#)" *Journal of Thermoplastic Composite Materials* (Nov 2025).

EDUCATION

Bachelor's Degree in Materials Engineering

Ain Shams University | University of East London | Graduated: 2025 | GPA: 3.17/ 4.0

Graduation Projects:

- Feasibility of Cladding Pure Copper on Aluminum 6082 Using Wire Arc Additive Manufacturing (WAAM)
- Process Optimization for Physio-mechanical Behavior of 3D-Printed PETG and PETG-CF for Automotive Applications

LANGUAGES

- Arabic: Native
- English: Professional Working Proficiency